

Full-Signal Comparison of Independent PN-IQ, GMP, and Sparse PNNN

Objective and protocol. The comparison uses the same modeled-block X/Y samples for every model. The classic selector provides 19,661 identification rows (approximately 4% of the capture); an internal 16,711/2,950 split selects regularization and neural epochs. Every final model is fitted on all identification rows and evaluated on all 491,520 signal rows. The full-signal evaluation includes the identification samples and is not an independent generalization test.

Main results. The three principal parameter-matched models use 358 active real parameters; the no-PN control uses 394. NMSE is $10 \log_{10}(\sum |y - \hat{y}|^2 / \sum |y|^2)$.

Model	Real parameters	Full-signal NMSE (dB)	FLOPs/sample	Additional operations/sample
Independent PN-IQ	358	-38.534	1062	12 complex magnitudes, 2 divisions
Independent I/Q without PN	394	-37.810	1086	12 complex magnitudes
Parameter-matched GMP	358	-38.170	1899	14 complex magnitudes
Sparse PNNN N12	358	-35.900	876	14 complex magnitudes, 2 divisions, 12 ELU

The sparse-PNNN FLOP count assumes that zero weights are skipped. If the original full matrices are executed, the cost is 2252 FLOPs/sample. A complex magnitude includes one square root.

Complex-equivalent reference. Separately, the 100-term complex GMP baseline and its coupled PN-IQ real representation remain numerically equivalent. This equivalence statement does not refer to the 179-term parameter-matched GMP in the table.

Quantitative conclusions. Independent PN-IQ and the parameter-matched GMP both use 358 real parameters. PN-IQ improves full-signal NMSE by approximately 0.364 dB while requiring 837 fewer FLOPs/sample. Sparse PNNN N12 has the lowest analytical count when zero weights are skipped (876 FLOPs/sample), but it is approximately 2.634 dB behind PN-IQ and adds 12 ELU activations per sample.

Independent PN-IQ and Independent I/Q without PN both remove complex coupling and fit two independent real regressions; only the former uses phase normalization. They are not perfectly parameter-matched: the no-PN control has 394 parameters, 36 more than PN-IQ. Despite this larger budget, it is approximately 0.723 dB worse in full-signal NMSE and requires 24 more FLOPs/sample. For this capture, the result indicates that the gain is not explained by independent I/Q freedom alone and that the PN coordinate system also contributes; it is not a universal causal claim.

Thus, PN-IQ wins on NMSE, sparse PNNN wins on the stated sparse arithmetic count, and the GMP and no-PN control avoid the 12 ELU evaluations introduced by the sparse neural model. These analytical FLOPs do not imply a direct conclusion about latency, power consumption, or hardware resources.